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The Discontinuity Stability Analysis of Sub-Seismic Modelled Fractures

A. H. Alhasan and Y. A. Qahtani, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

Fracture modeling is a developing approach to calculate fracture porosity and permeability and has become a critical tool to help assess natural fractures and their stress state. In this study, image-log based fracture interpretation and fracture modeling methods were used to assess the stress regime, evaluate fracture network types, investigate the fractures' effect on porosity and permeability, and analyze the stresses affecting the modelled fractures to identify which fractures are conductive, and which are sealed.

Image-log based fractures are characterized looking for specific orientations from which the maximum and minimum horizontal stresses are implied. Moreover, the fractures are then modelled utilizing various petrophysical modelling approaches such as Kriging or Gaussian Random Function Simulation. After that, different fracture networks are generated to account for different scenarios, which are then upscaled using the Oda approach to produce porosity and permeability. Furthermore, fractures modelled in this study were then subject to the discontinuity stability analysis in order to analyze their present-day stress status, and identify which of these modelled fractures is critically stressed, and which is stable.

Image-log based fracture interpretation orientations and trends helped shed the light on the direction and orientation of the minimum and maximum horizontal stresses. It was implied that the minimum horizontal stress is parallel to any induced fractures. In addition, the generated fracture networks were upscaled to generate fracture permeability and fracture porosity maps via the Oda upscaling method, which utilizes fracture aperture, the volume of grid cells, and the Gaussian Random Function Simulation in order to predict the results at the given fracture location. Wherever fractures are oriented parallel to the maximum horizontal stress, fracture porosity and fracture permeability maps illustrate a higher value. Finally, the discontinuity stability analysis utilized elastic parameters and principal stresses to identify which fractures are conductive. For instance, on the one hand, fractures parallel to the maximum horizontal stress were mostly critically stressed or conductive similar to the observations made with fracture porosity and fracture permeability maps. This is because the maximum stress acts to keep these fractures open. On the other hand, fracture perpendicular to it are observed to be stable.

This work has furthered the understanding of fracture porosity, fracture permeability, and the stresses acting on these fractures within the area of study and has provided a computationally efficient, validated approach to modeling fracture porosity and permeability in areas with robust image log data sets. This

approach is broadly applicable and can be used in conjunction with dual porosity and dual permeability modeling to create integrated geo-models to assess other areas of interest.

Introduction

The discontinuity stability analysis of fractures significantly assists in enhancing the understanding of how present-day stresses affect the stability of sub-seismic modelled fractures. Today, this analysis is developing year by year to be one of the most crucial parts of a risking analyses in which fracture porosity and permeability play an important role in risking the reservoir seal capacity. This analysis attempts to differentiate between fractures that are conductive and fractures that might be sealing (Hennings *et al.,* 2012). Also, this study would act as a crucial part of the risking strategy in terms of analyzing migration pathways to the target reservoir, and whether the reservoir is in communication with an aquifer for instance, that might introduce water and contaminate the reservoir. On the one hand, the fractures could highly enhance the permeability thus creating connectivity in the reservoir which improves direct hydrocarbon extraction or, in unconventional reservoirs, hydraulic fracturing. On the other hand, it might act as a bridge between a reservoir and an aquifer thus causing water encroachment. Therefore, the study would improve accuracy in terms of risking potential prospects when it comes to seal breach, and it would improve the understanding of the reservoir top seal and lateral seal capacity.

The discontinuity stability analysis tends to define fractures that are conductive using principal stresses acting on the area of interest in this study shown in Figure 1 below in a proglacial sand channel reservoir. In addition, the analysis requires a fracture model with a Discrete Fracture Network (DFN). The reason behind this is that the modelled fractures in the DFN are generated as individual sub-seismic fracture planes, where the present-day principal stresses are applied to them to analyze their individual stress state. The fractures in the area of interest displayed below in Figure 1 are modelled utilizing image log fracture interpretations and other inputs such as principal stresses and seismic attributes as fracture drivers. Eventually, the study aims to distinguish between fractures that are stable (sealing) and fractures that are stressed (conductive). The paper attempts to showcase the methodology and workflow that this analysis has followed from interpreting and modelling these fractures in proglacial tight sand channels to applying the stresses for the discontinuity stability analysis where the fractures are separated into sealing or conductive. Moreover, the results will be discussed highlighting accuracy and precision of the fracture model, its properties such as fracture porosity and permeability, and the stability analysis. Finally, the paper will conclude with novelty of work and recommendations after a discussion of the analysis results.

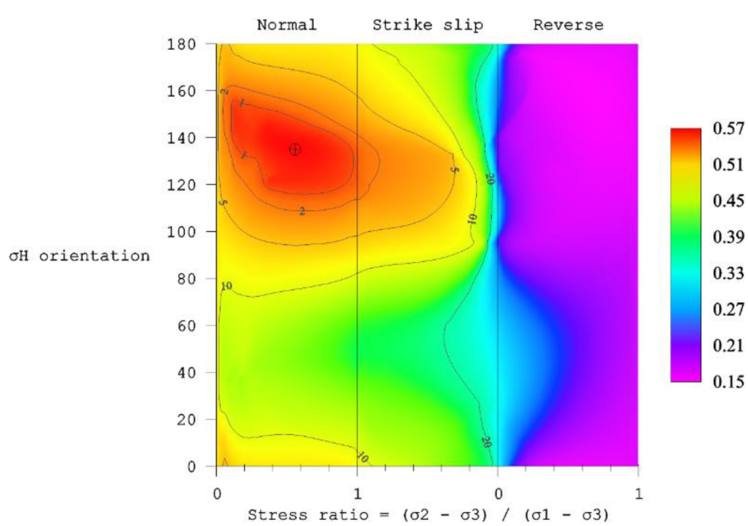

Figure 1—Tectonic model showcasing stress regimes affecting study area

Methodology

It is of a substantial impact to enhance and elevate the knowledge of fractures and their stress state in the subsurface. Specifically, sub-seismic fractures that are interpreted on the image logs coming from vertical wells in the area. First of all, faults were manually picked on the seismic as interpretation sticks to be used later on in the modelling stage, and to give an implication regarding structures or compartments in the area of interest. Here, several seismic-based volumetric attributes were generated to assist with discontinuity detection such as edge enhancement, which focuses on and highlights discontinuities that may be confused with other features on the seismic. This method succeeded in illustrating the discontinuities (faults and fractures) much better than the traditional seismic only approach after applying structural smoothing. Also, dikes were mapped from potential field data. Specifically, using the magnetics maps with various frequency ranges to account for major and minor igneous intrusions. Then, fractures were manually picked on image logs and characterized based on their observed type whether open or closed.

Having interpreted faults from seismic, igneous intrusions from potential field data, and fractures from image logs, the next step would be generating a tectonic model to imply the regional stresses affecting the area of interest of this study. Three scenarios were accounted for, beginning with including all faults, fractures, and dikes. The second scenario was modelling tectonic stress regimes using only faults and fractures, and the third and final scenario was a model including only dikes. Scenario 1 was selected as it was the most representative of the date and stress regimes affecting the area. Figure 1 below showcases that the area of interest is mostly undergoing normal and strike slip regimes with negligible reverse faulting evidence. The reason behind generating a tectonic model is that it could assist in generating some fracture properties such as dip, azimuth, and density, which are going to be important inputs to assist generating a more precise and accurate DFN.

Moving on, all of the interpretations were picked in time domain, hence, they required a velocity model to be able to convert them to depth to create the structural model and geological grid. After that, the geomechanical model in the area was utilized to extract principal stresses and fracture stress parameters. These were used to model the fractures deterministically and stochastically. The deterministic model relied mainly on interpreted faults and dikes and modelled fracture swarms surrounding them. The stochastic model covered the entire area study, which was preferable for the purpose of this analysis. The fractures modelled were then calibrated with image log interpreted fractures and manually interpreted faults/fractures on the seismic. After selecting the most representative DFN model, which was mainly driven by seismic attributes as well as fault/dike damage zones, it was upscaled using the Oda method to generate important fracture properties such as fracture porosity and fracture permeability by relating parallel fractures to permeability (Chen *et al.,* 2015). These properties should correlate with the discontinuity stability analysis where high permeability, for instance, would imply a high density of conductive fractures. Though, it should be noted that vertical fracture density as an input might incur some uncertainties when stochastically modelling the DFN. Eventually, the geomechanical model containing principal and elastic stress parameters was used as the main input as well as the DFN model created in a discontinuity stability analysis tool. This tool would estimate the amount of the stress acting on those modelled fractures, and take into consideration with that the orientation of the fractures to evaluate whether these fractures are critically stressed (conductive) or stable (sealed). Since the DFN was only modelled at the reservoir level, the resolution of the model was to be taken into consideration to sustain quality, accuracy, and precision. An important thing to note at this stage was the overburden, since this study is highly dependent on the depth of modelled fractures. This means that the overburden would affect the amount of stress the modelled fractures would need to reach a critical state and be conductive. In other words, the deeper the fracture, the more stress it requires to be conductive. According to experimental work, the normal stress was correlated inversely with fracture permeability (Cardona *et al.,* 2021). In other words, the higher the stress is, the lower the fracture permeability should be. Finally, the results of this study were validated and calibrated from existing data such as drilled wells. This could be done swiftly by looking at drilling events, and connecting any drilling issues to stressed fractures. For example, if a well is drilled on a cluster of fractures that are conductive, loss of circulation is expected at the reservoir level where we have unstable fractures. This of course significantly affects reservoir quality, and would assist in risking potential future prospects.

Results and Discussion

The discontinuity stability analysis can help estimate fractures stress state and assign which are conductive and which are sealing. The generated DFN showcased how fractures were propagated in the area. Figure 2 below displays how these individual fracture planes show more density around interpreted faults and dikes. This is expected as the rock in observed deformation zones and major shear activity would fracture more. This is seen more clearly in reservoirs that are not shale/clay dominated.

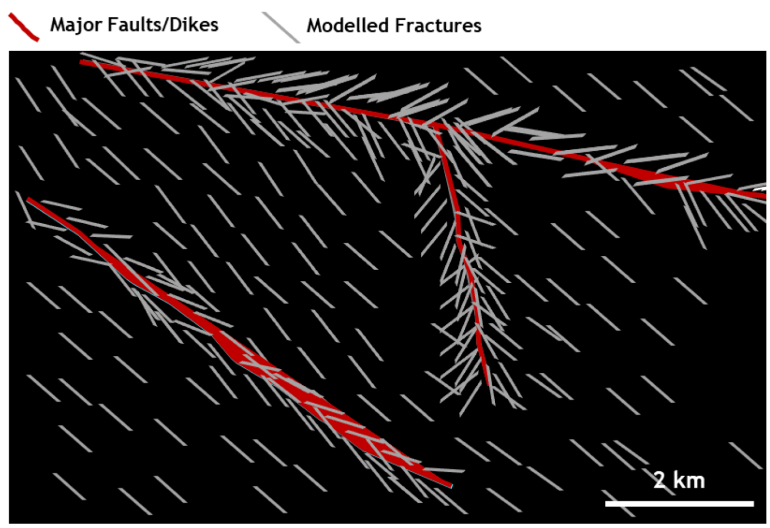

Figure 2—Discrete Fracture Network

An approach of calibrating the generated fracture model is to compare the DFN with manually interpreted fractures and image log fractures. Figure 3 clearly illustrates that they are all striking in the same orientation. This is one way of making sure that the modelled fractures follow the general trend in the area.

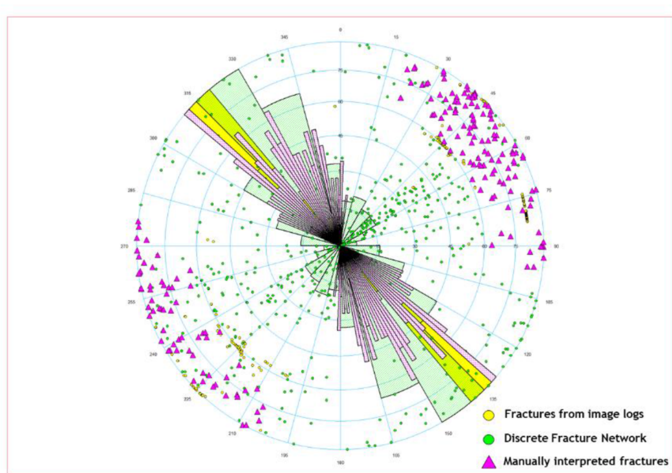

Figure 3—Stereonet to validate modelled fractures with interpreted fractures

After following the workflow illustrated in the methodology section, and validating the model by comparing it with manually interpreted fractures, the analysis resulted in the DFN color coded by which fractures are conductive and which fractures are stable or closed. For example, Figure 4 displays modelled fractures that are conductive (red). A very important observation on the figure is that almost all of the critically stressed fractures are parallel to the maximum horizontal stress. In this case, this adds to the accuracy of the analysis since it is expected that fractures which are parallel to the maximum horizontal stress are more prone to be open or conductive. This is because the maximum stress acts to keep them open. Olson *et al.* (2009) claims that with no shear failure, the right amount of stress could help keep fractures conductive. They also state that the permeabilities of said fractures are could be enhanced with the added stress. Moreover, it is claimed that fracture planes parallel to the maximum horizontal stress are not necessarily conductive or open. The could be closed down by other different factors affecting the area (Laubach *et al.,* 2004).

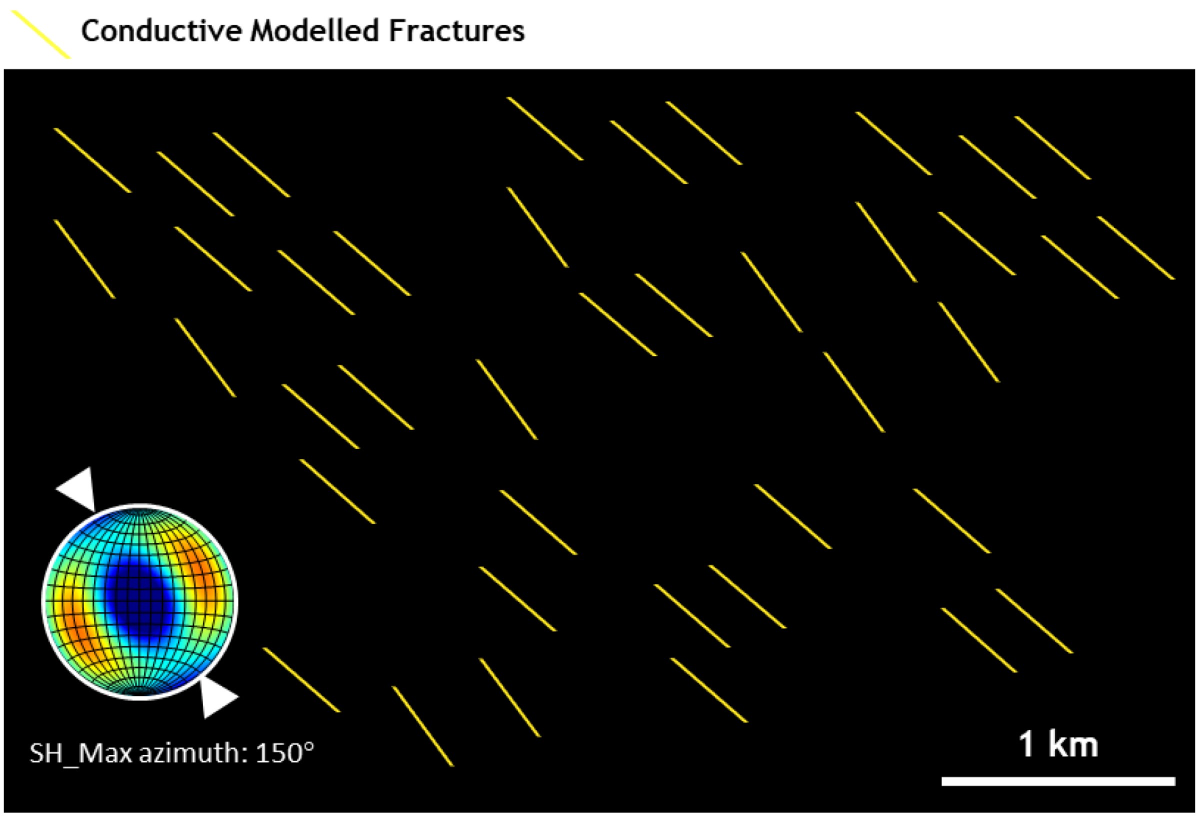

Figure 4—Filtered DFN showing conductive fractures are parallel to SHmax (around 150 degrees)

On the contrary, fractures that are parallel to the minimum horizontal stress, are mostly stable or closed due to the maximum stress acting from the opposite orientation to close them down. These results further support the validity of the analysis conducted in this study. The figure below shows fractures in green that are stable or sealed, and fractures in red that are conductive. It is clear from the figure that in most areas, where the fracture plane is perpendicular to SHmax, it would be closed down and sealed. An important note here is that one of the limitations of this analysis was the lack of sufficient control points in the southern part of the study area. It is to be taken into consideration when looking at that specific area. However, some of the observed and analyzed fracture planes in Figure 5 illustrate that there are some conductive fractures which are misaligned with respect to the maximum horizontal stress. These fractures could be close to faults, or they could be in regions of high pore pressure. This further supports what was claimed by Laubach *et al.,* (2004) previously.

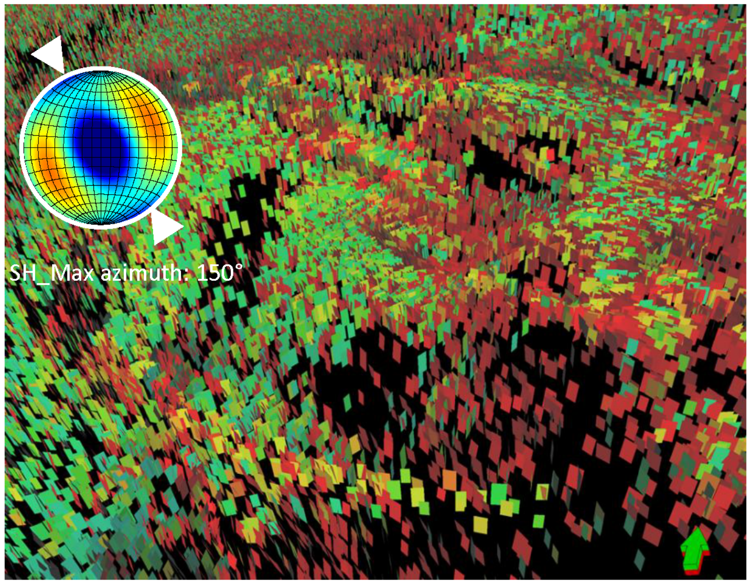

Figure 5—DFN color coded by conductive (red) and sealed (green) fractures

Furthermore, the analysis attempts to look at and investigate Mohr circle diagram to attempt to identify the amount of pore pressure each fracture in the modelled DFN requires to reach a critical state, in which the fracture is open and conductive. Figure 6 below displays Mohr circle diagram, which displays the shear failure envelope that indicates how much pore pressure is needed for each fracture in the DFN to reach failure. This pressure is commonly known as the pressure to fail. When applying additional pressure, these fractures would be driven to a critical state (being conductive). Any point that is tangent to the line or above it is at critical state or has failed. As can be observed from Figure 6, some fractures have already reached failure. This could be good or bad depending on the objective. On the one hand, these conductive fractures could cause loss of circulation. However, on the other hand, if targeted correctly, the conductive fractures could significantly enhance permeability and fluid flow.

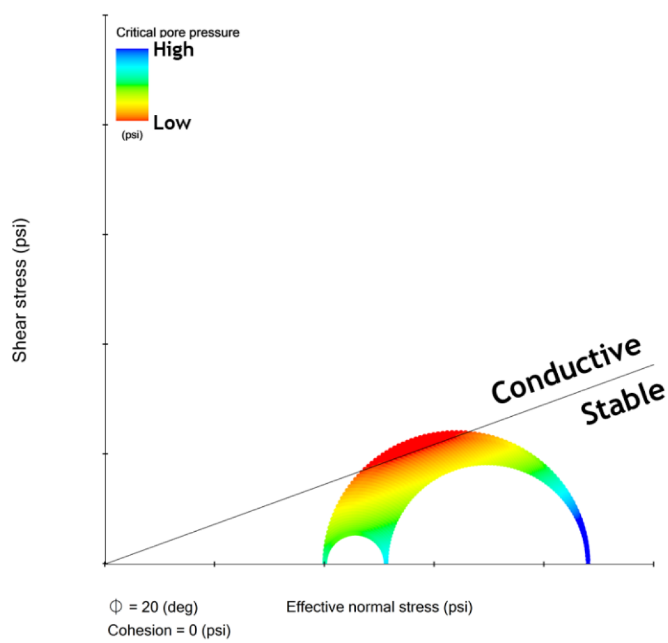

Figure 6—Mohr Diagram indicating how much pore pressure needed for fractures to be conductive

Finally, this study has furthered the risking strategy in the area by enhancing the chance of success in future prospect placement. For instance, conductive fractures, if encountered, could create drilling issues such as deformed casing or a crooked hole. In addition, conductive fractures could cause early water breakthrough if connected to both the reservoir and an aquifer. Nevertheless, if conductive fractures are encountered in a tight reservoir, and confined to it (not connected to an aquifer), then it could possibly enhance hydrocarbon recovery. Thus, the analysis would significantly help reduce encountering water if utilized in the risking strategy. According to Becker *et al.* (2019), one of the most crucial aspects of reservoir development and risk strategy enhancement is understanding how these fractures and their orientations would affect production and increase connectivity in tight reservoirs. Since the reservoir targeted in this study is a tight sand reservoir, this analysis would significantly improve the understanding of flow pathways and connectivity.

Conclusions

To conclude, the purpose behind this study was to interpret fractures manually from image logs and faults from volumetric attributes, as well as igneous intrusions from potential field maps. This has helped in generating a tectonic model that was used to estimate fracture properties like density, dip, and azimuth. In addition, the main objective of this analysis was to enhance the knowledge and understanding of how present-day principal stresses would affect interpreted and modelled fractures causing them to either stay open or close down. The results do in fact support what is expected. Meaning that fracture planes modelled in the DFN that are parallel to the maximum horizontal stress are conductive, and fractures that are perpendicular to the maximum horizontal stress are stable or sealed. The outcome of this project was validated utilizing real-time data and daily drilling reports. This was successfully completed by attempting to connect loss of circulation to dense clusters of conductive fractures. This type of analysis could be applied in areas where there are integrated datasets of interpreted faults, fractures, and igneous intrusions to assess the top and lateral seal capacity. To sum up, this analysis could be improved upon further by integrating the geomechanical model to populate the fractures in the area using principal stresses rather than a stochastic model. Fortunately, today, there are a lot of information and workflows in the literature that support fracture modelling and the discontinuity stability analysis even though it is yet considered to be a newly found approach. Finally, an approach to minimize inaccuracy and imprecision in the analysis is to address the limitations mentioned in the study. This would definitely enhance the outcome of the analysis and elevate and solidify the understanding of the area.

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